Exercise 7. Prove that {a,b} and {gcd(a,b)} have the same set of common factors.

Let g = gcd(a,b). By definition, g | a and g | b.

If d is a common factor of a and b, then d | a and d | b, so d divides every common divisor, hence d | g. Conversely, if d | g, then since g | a and g | b, transitivity gives us d | a and d | b, so d is a common factor of a and b.

Thus, the set of common factors of a and b equals the set of divisors of g.

Exercise 14. Prove that for m != 0, (a mod m) ≡ a (mod m), where the "mod" on the left is the binary operator, while the "mod" on the right is the relation.

Definition 13. Given a and b with $b \neq 0$, we define $a \bmod b$ as:

$$a \bmod b := a - \left\lfloor \frac{a}{b} \right\rfloor b.$$

By Definition 13, a mod m = a - floor(a/m)m. Let q = floor(a/m). Then, a mod m = a - qm.
Thus, a - (a mod m) = qm, so m | (a - (a mod m)).
Hence, (a mod m) **≡ a (mod m)**

Exercise 19. Prove that the function always returns a prime factor of n, assuming n >= 2.

Assume d is the smallest divisor of n with d >= 2. Suppose, for contradiction, that d is composite. Then d = ab with 2 <= a < d and 2 <= b < d. Since a | d and d | n, we have a | d | n, a | n. But since a is a divisor of n with 2 <= a < d, this contradicts the minimality of d. Therefore, d cannot be composite, so d is prime.

Exercise 22. Prove that looping from 1 to n takes O(n) time.

Looping from 1 to n requires n iterations. Each iteration performs a constant amount of work O(1). Assuming constant-time operations per iteration, the total time is n * O(1) = O(n).

Exercise 23. Prove that if n >= 2 and n has no factors in [2, √n], then n is a prime.

Suppose, for contradiction, that n is composite. Then n = ab, with 2 <= a <= b. Then a <= √n, since a^2 <= ab = n. Thus a is a factor of n in [2, √n], contradicting the hypothesis. Therefore, n cannot be composite, so n is prime.

Exercise 24. Why did we not simply compute $\lfloor \sqrt{n} \rfloor$ and run the loop for p from 2 to $\lfloor \sqrt{n} \rfloor$?

idk man you tell me but i think we should.

The trial division loop only needs to check divisors up to √n because if n is composite, it has a factor <= √n by Exercise 23. The given code loops to n for simplicity of presentation,but checking only up to √n should yield the same correctness with improved efficiency.

Exercise 26. Prove that a is a gcd of a and 0. Note that this includes the case a = 0.

> **Definition 6.** We say g is a **greatest common divisor**, or **gcd**, of a and b if $g \mid a$, $g \mid b$, and for every d such that $d \mid a$ or $d \mid b$, it follows that $d \mid g$.

Let g = a. By Definition 6, g | a, and g | 0 because a | a and a | 0 since a * 0 = 0.
For any d such that d | a and d | 0, we have d | a = g.

Thus, a satisfies the definition of a gcd of a and 0. This holds for all integers including a = 0, where 0 is a gcd of 0 and 0.

Exercise 27. Prove that gcd(a,b) = gcd(b,a mod b).

Let r = a mod b, so by definition, a = bq + r with 0 <= r < |b|.

We show F{a,b} = F{b,r}.

(⊆) Let d | a and d | b, then d | a - bq = r, so d | b and d | r.
(⊇) Let d | b and d | r. Then d | bq + r = a, so d | a and d | b.

Thus, the common divisors are identical, so their greatest elements are equal. Hence, gcd(a,b) = gcd(b,r).

Exercise 28. Prove that Euclid's algorithm actually runs in $O(k^2)$ time.

Let a and b be positive integers with a >= b. Let the number of bits of a be k and of b be l (k >= l).

Each division step computes r = a mod b. Using grade-school division, this takes O(l(k - l + 1)) bit operations. The remainder r has at most l bits.

In consecutive steps, the number of bits of the arguments decreases rapidly. More precisely, if a >= b then a mod b < b, and often a mod b <= a /2. A standard analysis shows that the number

of steps is O(k) and each step's cost is proportional to the product of the bit–lengths of the operands.

Summing over all steps the total bit complexity is $O(k^2)$.

Exercise 30. Prove that if ax + by is a common divisor of a and b, then it has to be a greatest common divisor. (Hint: Prove that any common divisor of a and b divides ax + by).

Let d = ax + by, suppose d | a and d | b. Let g = gcd(a,b). Then g | a, and g | b so g | ax + by = d. Hence g <= |d| in absolute value.

Conversely, since d is a common divisor, d | g by definition of gcd. Thus, |d| <= g.

Therefore, |d| = g, so d is a gcd, up to sign.

Exercise 31. Suppose $l_1, l_2, \ldots, l_k$ are linear combinations of a and b and $c_1, c_2, \ldots, c_k$ are constants. Show that $c_1l_1 + c_2l_2 + \ldots + c_kl_k$ is also a linear combination of a and b.

Each $l_i = ax_i + by_i$ for some integers x_i and y_i, then

$$\sum_{i=1}^{k} c_i\ell_i = \sum_{i=1}^{k} c_i(ax_i + by_i) = a\left(\sum_{i=1}^{k} c_ix_i\right) + b\left(\sum_{i=1}^{k} c_iy_i\right).$$

Since $\sum c_ix_i$ and $\sum c_iy_i$ are integers, the result is a linear combination of a and b . ■

Exercise 32. Suppose a' = ax + by and b' = ax' + by'. Express a' mod b' as a linear combination of a and b.

By the division theorem, a' = qb' + r where r = a' mod b' and q = ⌊a '/b '⌋.
Then r = a'- qb' = (ax + by) - q(ax' + by') = a(x - qx') +b(y - qy')
Thus, a' mod b' = a(x - qx') + b(y - qy'), a linear combination of a and b.

Exercise 33. Prove that if the extended Euclidean algorithm returns (g,x,y), then g is a gcd of a and b, and g = ax + by.

[Base Case] When b' = 0, the algorithm returns (a', x, y). By Exercise 26, a' is a gcd of a' and 0. The invariant a' = ax + by holds by the initial call setup.

[Inductive Step] Assume the algorithm works for all inputs with fewer recursive calls. For inputs (a',x,y,b',x,',y') with b' != 0:

Let q = ⌊a ′/b ′⌋ and r = a' - qb'. The algorithm recursively calls extended-gcd(b',x',y',r,x-qx',y-qy').

By the induction hypothesis, the recursive call returns (g,x'',y'') such that:

- g is a gcd of b' and r
- g = b'x'' + ry''

By Exercise 27, gcd(b',r) = gcd(a',b'), so g is a gcd of a' and b'.

We substitute r = a' - qb'.

$$g = b'x'' + (a'-qb')y'' = a'y'' + b'(x'' - qy'')$$

The algorithm returns (g, x'',y'') as the result. We verify that these coefficients correctly express g in terms of the original a and b.

The invariant holds at the recursive call: b'=ax'+by' and r = a(x-qx')+b(y'qy') by Exercise 32. Substituting, we have:

$$g = (ax'+by')x'' + [a(x-qx')+b(y-qy')]y''$$
$$= a(x'x'' + (x-qx')y'') + b(y'x''+(y-qy')y'').$$

Thus, g = aX + bY for integers X,Y. The algorithm returns (g,x'',y'') where these coefficients satisfy the linear combination property. By induction, the claim must hold for all inputs.

Exercise 34. Prove Bezout's lemma using the correctness of the extended Euclidean algorithm.

Let a and b be two arbitrary integers. Run the extended Euclidean algorithm on (a,b). By Exercise 33, the algorithm returns (g,x,y) such that g = ax + by and g is a gcd of a and b. Thus, there exist integers x,y with g = ax + by for any gcd g of a and b.

Problem 36. Prove that if $g \nmid c$, then ax+by=c has no integer solutions.

Let g = gcd(a,b). Suppose ax+by=c has an integer solution (x,y). Since g | a and g | b, we have g | (ax + by) = c. Thus, g | c. By contrapositive, if $g \nmid c$, no solution exists.

Exercise 37. Prove that gcd(a,b,c) = gcd(gcd(a,b), c).

Let g = gcd(a,b,c) and h = gcd(gcd(a,b),c).

[=>] We have g | a, g | b, g | c, since g | a and g | b, we have g | gcd(a,b). Together with g | c, we get g | h.

[<=] h | gcd(a,b) and h | c. Since h | gcd(a,b), we have h | a and h | b, together with h | c, we get h | g.

Thus, g = ±h. Taking nonnegative gcds, gcd(a,b,c) = gcd(gcd(a,b),c).

Exercise 38. Solve the linear Diophantine equation in three variables: ax + by + cz = d for some constants a,b,c,d in the set of integers.

Let g = gcd(a,b,c). By Exercise 37, g = gcd(gcd(a,b), c).

1. If $g \nmid d$, no integer solutions exist, by Problem 36.
2. If g | d, we can write d = ge.

Solve gcd(a,b) = ax'+by' using extended Euclidean algorithm. Let h = gcd(a,b). Then h = ax'+by'.

Then hu+cv=g by extended Euclidean algorithm, giving g = hu+cv=(ax'+by')u + cv = a(x'v)+b(y'u)+cv

Thus, g = aX+bY+cZ where X=x'u,Y=y'u,Z=v. Multiplying by e, we have
d = ge = a(Xe)+b(Ye)+c(Ze). Therefore, a solution is (x,y,z) = (Xe,Ye,Ze).

Exercise 39. Solve the linear Diophantine equation in k variables: $a_1k_1 + a_2k_2 + \ldots + a_kx_k = c$ for some constants $a_1, a_2, \ldots, a_k$, c integers.

Let $g = \gcd(a_1, a_2, \ldots, a_k)$. By repeated application of Exercise 37, we have:

$$g = \gcd(\gcd(\ldots(\gcd(a_1,a_2),a_3), \ldots, a_k).$$

If $g \nmid c$, no integer solutions exist. If $g \mid c$, we can write $c = ge$.

Using the extended Euclidean algorithm iteratively:

- First, we find $g_2 = \gcd(a_1,a_2) = a_1u_1 + a_2u_2$.
- Then, $g_3 = \gcd(g_2,a_3) = g_2v_1+a_3v_2 = (a_1u_1+a_2u_2)v_1+a_3v_2=a_1(u_1v_1)+a_2(u_2v_1)+a_3v_2$.
- Continue inductively to get g = sum of all a_iw_i from i = 1,...,k.

Multiply by e to get c = ge = sum of all $a_i(w_ie)$.

Thus, a solution is $(x_1,x_2,\ldots,x_k) = (w_1e,w_2e,\ldots,w_ke)$.

Exercise 41. Prove that if $f : \mathbb{R} \to \mathbb{R}$ is nonnegative and monotonic (either non-decreasing or non-increasing) on $[a, b]$, then

$$\left| \int_a^b f(x)dx - \sum_{n=a}^{b} f(n) \right| \leq \max(f(a), f(b)).$$

Hint: If f is non-decreasing, prove that

$$\sum_{n=a}^{b-1} f(n) \leq \int_a^b f(x)dx \leq \sum_{n=a+1}^{b} f(n).$$

Find something similar if f is non-increasing.

Assume f is non-deceasing, with the non-increasing case being symmetric. Then for each integer with a <= n <= b - 1,

$$\int_n^{n+1} f(n)\,dx \leq \int_n^{n+1} f(x)\,dx \leq \int_n^{n+1} f(n+1)\,dx.$$

Thus, f(n) <= S^(n+1)_(n) f(x) dx <= f(n+1).

Summing from n = ceil(a) to floor(b) - 1 and adding the boundary adjustments, the total difference between the total integral and the sum is bounded by the first and last terms. Formally,

$$\sum_{n=\lceil a \rceil}^{\lfloor b \rfloor} f(n) - \int_a^b f(x)\,dx \le f(b) - f(a)$$

Since f is nonnegative, the absolute difference is at most max(f(a), f(b)).

Exercise 42. Prove that we can begin the inner loop at d^2 instead of 2d. In this variation, prove that the algorithm still takes Θ(mloglogm) steps.

If n is composite and has a prime factor d, then n = dk with k >= d when d <= √n. Thus, the smallest prime factor d will mark n when the loop reaches d and starts at d^2. Multiples smaller than d^2 (i.e, 2d, 3d, … d(d-1) already have a smaller prime factor and would have been marked earlier.

So the work becomes

$$\sum_{d \le \sqrt{m}} \left(\frac{m}{d} - d\right) \approx \sum_{d=2}^{\sqrt{m}} \frac{m}{d} = \Theta(m \log \log m)$$

The subtracted sum of all d is Θ(m), which is negligible. Thus, the complexity remains Θ(mloglogm).

Exercise 44. Prove that the total lengths of all lists in divs is <= mlogm + O(m).

For each d from 1 to m, the inner loop appends d to floor(m/d) multiples. The total number of appends is:

$$\sum_{d=1}^{m} \left\lfloor \frac{m}{d} \right\rfloor \le \sum_{d=1}^{m} \frac{m}{d} = m \sum_{d=1}^{m} \frac{1}{d}.$$

Using the harmonic series bound $\sum_{d=1}^{m} \frac{1}{d} \le \ln m + 1$,

$$\sum_{d=1}^{m} \left\lfloor \frac{m}{d} \right\rfloor \le m(\ln m + 1) = m \log m + \mathcal{O}(m).$$

Thus, the total list length is <= m log m + O(m).

Exercise 45. Suppose you have a valid value of pf[n] for each n from 2 to m. Explain how to commute the prime factorization of any n from 1 to m, such that the running time is proportional to the number of prime factors of n.

For n = 1, return an empty factorization. For any n >= 2, we follow this algorithm:

1. Let p = pf[n].
2. Initialize count k = 0.
3. While n mod p = 0:
    a. Increment k
    b. Set n = n/p
4. Output factor p^k
5. Recursively factor the remaining n (if n > 1)

Each division reduces n by at least one prime factor, so the total number of operations is O(number of prime factors with multiplicity).

Exercise 46. Prove that any rational number can be expressed uniquely as a quotient a/b of integers where a and b are coprime, and b is positive.

Let r be a rational number. Write r = p/q, with q != 0.

Let g = gcd(p,q). Set a = p/g and b = q/g. Then gcd(a,b)=1 and r=a/b.

If b < 0, multiply the numerator and denominator by -1 to make b > 0.

We show uniqueness. Suppose r = a/b = a'/b' with b, b' > 0 and gcd(a,b) = gcd(a',b') = 1. Then ab'=a'b. Since b | a'b and gcd(a,b)=1, we have b | b'. Similarly, b' | b. Thus, b = b' and consequently, a = a'.

Exercise 47. Prove that Q(a,b) + Q(a',b') = Q(a * (b'/g) + a' * (b/g), l), where g = gcd(b,b') and l = lcm(b,b').

Theorem 91. If $a, b \in \mathbb{Z}$ are nonnegative, then $\gcd(a,b)\,\mathrm{lcm}(a,b) = ab$.

Let l = bb'/g by Theorem 91. Then,

$$\frac{a}{b} + \frac{a'}{b'} = \frac{ab' + a'b}{bb'} = \frac{a(b'/g) + a'(b/g)}{bb'/g} = \frac{a(b'/g) + a'(b/g)}{\ell}$$

Since gcd(b/g,b'/g)=1, the numerator and denominator may share a common factor. However, when constructing Q, the representation is reduced to lowest terms. The expression shows the sum with denominator l, which is the least common denominator.

Exercise 49. Prove that xy mod m = (x mod m)(y mod m) mod m.

Let $x = q_x m + r_x$ and $y = q_y m + r_y$ where $r_x = x \bmod m$ and $r_y = y \bmod m$, so $0 <= r_x, r_y < |m|$. Then,

$$xy = (q_x m + r_x)(q_y m + r_y) = q_x q_y m^2 + q_x r_y m + q_y r_x n + r_x r_y$$
$$xy - r_x r_y = m(q_x q_y m + q_x r_y + q_y r_x), \text{ so } m \mid (xy - r_x r_y)$$
$$xy \equiv r_x r_y \pmod m.$$

By definition of the binary mod operator,

$$(x \bmod m)(y \bmod m) \bmod m = (xy) \bmod m$$

Note that there is a single recursive call to exponentiation here! (Note that y^2 can be computed as $y \cdot y$.) We can be explicit about this, so we can see that there is indeed a single recursive call:

$$\mathsf{pow}(x, n) = \begin{cases} 1 & \text{if } n = 0 \\ \mathsf{pow}(x, n-1) \cdot x & \text{if } n > 0 \text{ and } n \text{ odd} \\ \mathsf{pow}(x, n/2)^2 & \text{if } n > 0 \text{ and } n \text{ even.} \end{cases}$$

Exercise 50. Prove that this takes $\mathcal{O}(\lg n)$ steps.

Let T(n) be the number of recursive calls for exponent n. If n = 0, T(0) = 1. If n is odd, T(n) = T(n-1)+1. If n is even, T(n) = T(n/2)+1.

In the odd case, n-1 is even, so T(n) = T((n-1)/2) + 2. Thus, in both cases, T(n) <= T(⌊n/2⌋)+2, which gives us T(n) = O(log n).

Exercise 55. Suppose $a_1, \ldots, a_k, m_1, \ldots, m_k \in \mathbb{Z}$ and the m_is are pairwise coprime, i.e., $\gcd(m_i, m_j) = 1$ for 1 ,= i, j <= k with i != j. Prove that there exists a solution x to the congruences:

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k}, \end{aligned}$$

and that this solution is unique modulo $m_1 m_2 \ldots m_k$.

We prove by induction on k.

[Base Case] Let k = 2. This is the Chinese Remainder Theorem.
[Inductive Step] Assume the system for k - 1 congruences has a solution x_0 unique modulo $M = m_1m_2\ldots m_{k-1}$. We need x satisfying

$$x \equiv x_0 \pmod{M},\ x \equiv a_k \pmod{m_k}$$

Since $\gcd(M, m_k) = 1$, by pairwise coprimality, the base case guarantees a solution x modulo Mm_k, unique modulo Mm_k.

Thus, a solution exists for k congruences, and uniqueness modulo the product follows by induction.

Exercise 59. Prove that -1 and 1 are units
1 | 1 because 1 * 1 = 1.
-1 | 1 because (-1)(-1) = 1.
Thus, both divide 1, so they are units.

Exercise 60. Prove 0 is not a unit, unless 1 = 0.
If 0 | 1, then there exists an integer c such that 0 * c = 1. In a nontrivial ring, 1 cannot be 0, so 0 cannot be a unit.

Exercise 61. Prove if x,y, are units, then xy is a unit.
Since x | 1, there exists an integer a with xa = 1. Since y | 1, there exists a b such that yb = 1. Then (xy)(ab) = x(yb)a = x * 1 * a = xa = 1. Hence, xy | 1, so xy is a unit.

Exercise 62. Prove that if x is a unit, then x | y for any y.
Since x is a unit, there exists a u with xu = 1. Then y = y * 1 = y(xu) = x(yu), so x | y.

Exercise 63. Prove if y is a unit and x | y, then x is a unit.
Since x | y, there exists an a such that xa = y. Since y is a unit, there exists yb = 1. Then, x(ab) = (xa)b = yb = 1, so x | 1, therefore, x is a unit.

Exercise 65. Prove that if x, d, u is in the set of integers such that u is a unit, then d | x if and only if d | ux.

[=>] If d | x, then x = dk for some k. Then ux = u(dk) = d(uk), so d | ux.
[<=] If d | ux, then ux = dl for some l. Since u is a unit, then there exists v with uv = 1. Multiply both sides by v, x = v(ux) = v(dl) = d(vl), so d | x.

Exercise 72. Prove that if p, q (integers) are primes such that p | q, then p = uq for some unit u.

Since p | q, q = pk for some k. Because q is prime, it is irreducible. Thus, in q = pk, one factor must be a unit. If k is a unit, then $p = k^{-1}q$ is a unit multiple of q. If p were a unit, then p | 1 implies p is in {1, -1}, which is not prime. Hence, p = uq with $u = kq^{-1}$ as a unit.

Exercise 73. Prove that if p ∈ ℤ is prime and u ∈ ℤ is a unit, then up is prime.

Let up | xy. Then $p \mid u^{-1}xy$ since multiplying by the unit u^{-1} preserves divisibility. Because p is prime, $p \mid u^{-1}x$ or p | y. If $p \mid u^{-1}x$, then up | x, again by multiplying by u. Hence, up | x or up | y. Also, up != 0 and up is not a unit, otherwise, p would be a unit. Thus, up is prime.

Exercise 75. Prove that if n ∈ ℤ is composite, and u ∈ ℤ is a unit, then un is composite.

Since n is composite, by Definition 74, m is not zero, not a unit, and not prime. Moreover, n can be written as n = ab, where a and b are nonzero nonunits. Then un = (ua)b.

Because u is a unit, ua is also a nonzero nonunit,)if ua were a unit, then multiplying by u^{-1} would make a unit, which is a contradiction). Thus, un is a product of two nonzero nonunits, so un is reducible and therefore composite (nonzero, nonunit, not prime).

Exercise 77. Prove that if g is a gcd of a and b and u is a unit, then ug is also a gcd of a and b.

Since g | a and g | b, we have ug | a and ug | b. Let d be any common divisor of a and b. Then, d | g because g is a gcd. Since d | g and u is a unit, d | ug (multiply g = dk by u). Thus, ug satisfies the definition of a gcd.

Exercise 78. Prove that if g and g' are gcds of a and b, then g = ug' for some unit u.

Since g is a gcd, g' | g because g' is a common divisor. Similarly, g | g'. Thus, g = g' * k and g' = g * l for some integers k and l. Substituting, we have g = glk, so g(1 - lk) = 0. If g != 0, then lk = 1, so k and l are units. If g = 0, then a = 0, b= 0. And any gcd is 0, so g = g' = 0 = 1*0. Hence, g = ug'.

Exercise 84. Prove that n is prime if and only if it has exactly one prime in its prime factorization.

[=>] If n is prime, then by the Fundamental Theorem of Arithmetic, $n = up_1p_2\ldots p_k$ where u is a unit and each p_i is a positive prime. Since n is a prime, it is irreducible. Thus, k = 1 and $up_1 = n$. Because n and p_1 are positive primes, u = 1. Hence, exactly one prime appears.

[<=] If the prime factorization of n contains exactly one positive prime, say n = up with unit u and positive prime p, then by Exercise 73, up is prime. Hence, n is prime.

Exercise 89. Let a,b be integers. Prove that if l, an integer, is an lcm of a and b, and u is a unit, then ul is also an lcm of a and b.

Since l is a common multiple, a | l and b | l. Multiplying by the unit u, or using Lemma 64, a | ul and b | ul.

Now, let m be any common multiple of a and b. Then l | m because l is an lcm. Since l | m, we have m = lk. Then, ul | m because $m = (ul)(u^{-1}k)$. Thus, ul satisfies the definition of an LCM.